30 June 1955
SUPERSEDING
NAVORD OS 3064I
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MILITARY SPECIFICATION

PISTOL, DEPTH CHARGE, MARK 6 MOD 2

1. SCOPE

1.1 The manufacture, assembly, and preparation for delivery of the Depth Charge Pistol Mk 6 Mod 2 are governed by this specification and all other items on the applicable list of drawings. In addition, this specification prescribes methods of inspection and tests which comprise in part the procedures upon which product acceptance will be based. The Depth Charge Pistol Mk 6 Mod 2 is an adjustable hydrostatic depth charge pistol with spring adjusted depth setting ranges of 30 to 300 feet and valve controlled 350 to 1000 feet.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-C-16173 -- Corrosion Preventive, Solvent Cutback, Cold-Application.

STANDARDS

MILITARY

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

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DRAWINGS

BUREAU OF ORDNANCE

SK No. 109321 -- Depth Charge Pistol Mk 6 Mod 2, List of Drawings, and all documents listed on the above LD.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 General.

3.1.1 Depth Charge Pistols Mk 6 Mod 2 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings SK No. 109321 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Springs. -- Particular care shall be exercised in the application of rust proofing processes, specified on the spring drawings, in order that the physical properties of the springs may not be adversely affected.

3.3 Rust preventive compound. -- All exposed metal parts shall be coated with rust preventive compound complying with Specification MIL-C-16173 Grade II. This coating shall be applied by dipping or spraying and surplus liquid shall be drained off or blown off with compressed air.

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3.4 Carrying flange, bellows, and stem subassembly. -- The Carrying Flange, Bellows, and Stem Subassembly, Drawing 433977, referenced on Sketch No. 109321, shall withstand, without leakage, an internal air pressure of 50 pounds per square inch when tested as specified in 4.4.1.1.

3.5 Depth setting mechanism for deep firing.

3.5.1 Valve opening pressure (1000 foot setting). -- With the depth setting pointer at the 1000 foot mark, the valve shall open at a pressure between 415 and 445 pounds per square inch when tested as specified in 4.4.1.2.1. The valve shall not vibrate or chatter at the point of opening within the required range of pressures.

3.5.2 Leakage (1000 foot setting). -- The Depth Setting Mechanism for Deep Firing, Drawing 432885, referenced on Sketch No. 109321, with the depth setting pointer at the 1000 foot mark, shall withstand, without leakage, an internal water pressure ranging from 0 to 415 pounds per square inch when tested as specified in 4.4.1.2.2.

3.5.3 Valve opening pressure (400 foot setting). -- With the depth setting pointer at the 400 foot mark, the valve shall open at a pressure between 150 and 180 pounds per square inch when tested as specified in 4.4.1.2.3.

3.5.4 Valve closing pressure (400 foot setting).-- After opening at the pressure specified in 3.5.3, gradual reduction of pressure shall cause the valve to close at a pressure between 46 and 90 pounds per square inch below the opening pressure when tested as specified in 4.4.1.2.4.

3.5.4.1 Tagging. -- After completion of the valve closing test as specified in 4.4.1.2.4, a tag shall be attached to the depth setting mechanism. The following information shall be recorded on the tag:

"_____Opening Pressure at 400 Foot Setting.
_____Closing Pressure at 400 Foot Setting."

3.5.5 Leakage (400 foot setting). -- After completion of the valve closing test specified in 4.4.1.2.4, application of 120 pounds per square inch pressure shall not cause leakage at a rate in excess of five cubic inches (82cc) per minute when tested as specified in 4.4.1.2.5.

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3.6 Calibration.

3.6.1 Depth setting mechanism for deep firing. -- The depth setting mechanism for deep firing shall be calibrated as specified in 3.6.1.1 through 3.6.1.6.

3.6.1.1 The depth setting pointer shall be turned to the 700 foot mark and the mechanism placed in the fixture shown on Drawing 423550, referenced on Sketch No. 109321. Precautions shall be taken to see that all calibration apparatus valves in the direct waterline to the test fixture are wide open during this test since pipe restrictions may cause valve chattering or vibration.

3.6.1.2 Water pressure shall be applied and shall be increased at the rate of approximately 11 pounds per square inch per second until the valve opens wide. This can be determined by a sudden rush of water as compared with a slight trickle which occurs at a few pounds before the actual opening point. The opening pressure shall be observed. Then the pressure shall be reduced until the valve closes. Complete closing of the valve will be insured by reducing the pressure 50 pounds per square inch below the point where most of the water flow suddenly ceases.

3.6.1.3 The valve shall open within the range of 290 to 315 pounds per square inch. The valve spring tension shall be controlled by turning the adjusting screw to bring the valve opening pressure within the range noted while repeating the cycle of pressure application and release.

3.6.1.4 When an opening pressure within the required range has been obtained, two cycles of pressure application and release shall be made without disturbing the position of the adjusting screw. These two valve operating cycles shall be made to check the satisfactory opening pressure initially obtained.

3.6.1.5 During the second valve operating cycle specified in 3.6.1.4, the mechanism shall be tested for leakage in accordance with 4.4.2.

3.6.1.6 When a depth setting mechanism for deep firing has been satisfactorily calibrated to function within the required limits, the mechanism shall be carefully removed from the testing fixture in order not to disturb the position of the spring adjusting screw. As soon as

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practicable, the adjusting screw shall be locked in place by soldering, as noted on Drawing 433979, referenced on Sketch No. 109321, to prevent any possibility of change in calibration.

3.6.1.7 After the spring adjusting screw has been locked in place with solder, the index pointer of the depth setting mechanism shall be registered to the column of the letter "T" in the word "FEET" engraved on the dial plate. The retainer nut shall then be screwed down on the spring retainer until it is finger tight against the spring adjusting screw. Precautions must be taken to insure that the contact surface between the spring adjusting screw and retainer nut is clean and free of any foreign substance. The retainer nut shall be secured to the spring retainer using a small torch and fine wire solder. This operation shall be performed with the mechanism immersed in water as noted on Drawing No. 433979 and with due care to avoid solder getting into the hole in the spring retainer or holes and slots in the spring adjusting screw.

3.6.1.8 Satisfactorily calibrated depth setting mechanisms with soldered spring adjusting screws and spring retainer nuts shall be made up with associated parts to form the subassembly shown on Drawing No. 432885, referenced on Sketch No. 109321. A torque wrench shall be used to screw the depth setting mechanism in place. The torque applied to assemble the mechanism shall be not less than 60 lb-ft nor more than 75 lb-ft. During subsequent tests, repeated torque application shall not be used to bring valve opening within required limits if failure is recorded on initial test.

3.6.2 Depth setting 30 feet to 300 feet. -- The pistol shall be calibrated and adjusted as specified in 3.6.2.1 through 3.6.2.4.

3.6.2.1 Mount the pistol in the fixture shown on Drawing No. 437961, referenced on Sketch No. 109321.

3.6.2.2 Back off the locking spring and remove the detonator holder and collar to allow a clear passage for the firing plunger which is ejected when the pistol is fired. Position a padded container to catch the plunger.

3.6.2.3 With the air supply attachment engaging the pistol flange outside of the dial lock ring, the index

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pointer set at "100", and the two index pointer securing screws centrally located in their slots, apply air pressure gradually to determine the pressure required to fire the pistol. Should the firing pressure be less than 43 or greater than 46 pounds per square inch, the mechanism shall be adjusted as specified in 3.6.2.4 as often as necessary to bring the firing pressure within these limits.

3.6.2.4 Adjustment of Index pointer.

3.6.2.4.1 Release the air pressure.

3.6.2.4.2 Using a brass rod with rounded ends reserved specifically for this purpose, depress the ball valve to permit entrapped air to escape from the hydrostatic chamber.

3.6.2.4.3 Using a recocking tool, replace the firing plunger in the pistol.

3.6.2.4.4 Loosen the two securing screws of the index pointer.

3.6.2.4.5 Keeping the index pointer in position at the "100" mark, rotate the depth setting sleeve a small amount. Rotation shall be clockwise if the firing pressure was less than 43 pounds, counterclockwise if greater than 46 pounds.

3.6.2.4.6 Tighten the index pointer securing screws.

3.6.2.4.7 Reapply air pressure to determine the new firing pressure.

3.7 Sealing. -- Upon completion of the calibrations specified in 3.6, the index pointer and the deep firing depth setting mechanism housing shall be locked in place by self locking set screws. In addition, an alignment mark shall be made with a small chisel across the edges of the index pointer discs, as indicated in the drawings, to insure that the correct adjustment is maintained.

3.8 "300" index pointer setting. -- The assembled pistol shall fire between 115 and 145 rounds per square inch pressure when tested as specified in 4.4.3.

3.9 Maximum pressure requirements. -- Depth Charge Pistols Mk 6 Mod 2, manufactured in accordance with this specification, shall conform to the following requirements when subjected

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to the maximum pressure tests specified in 4.4.5. Tests for these requirements shall be made in the order shown below. Prior to each test the calibration shall be tested as specified in 4.4.4.

3.9.1 400 foot setting. -- With the index pointer set at "100" and the depth setting mechanism for deep firing set at "400", the pistol shall fire when water pressure of between 155 and 205 pounds per square inch is applied.

3.9.2 700 foot setting. -- With the index pointer set at "100" and the depth setting mechanism for deep firing set at "700", the pistol shall fire when water pressure of between 280 and 330 pounds per square inch is applied.

3.9.3 1000 foot setting. -- With the index pointer set at "100" and the depth setting mechanism for deep firing set at "1000", the pistol shall fire when water pressure of between 415 and 465 pounds per square inch is applied.

3.9.4 0-300 foot setting. -- With the index pointer set at "SAFE" and with the depth setting mechanism for deep firing set at "0-to-300", the pistol shall not fire when subjected to hydraulic pressure of 1000 pounds per square inch. Rupture of the bellows at 250 pounds per square inch, minimum, during the pressure application shall be considered evidence of satisfactory performance under this test, provided that pressure application is continued beyond the point of bellows rupture to insure that the surge of fluid through the bellows rupture and into the mechanism casing does not operate the release plunger and cause firing of the pistol.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as

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determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of ten (10) Depth Charge Pistols Mk 6 Mod 2 including both plain and knobbed inlet valve covers, and two (2) spare pistol bellows shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These pistols and extra component parts shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.4 Tests.

4.4.1 Watertightness and valve operating pressures.

4.4.1.1 Carrying flange, bellows, and stem subassembly leakage test. -- With the unit so secured as to prevent longitudinal movement of the bellows, air pressure of 50 pounds per square inch shall be supplied to the interior through suitable fittings engaging the pistol carrying flange. There shall be no evidence of leakage under this

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pressure when the exterior surfaces and soldered joints of the subassembly are painted with soap suds solution or, as an alternative, when the subassembly is immersed in water. After this test, the parts shall be thoroughly cleaned and dried. Failure to pass this test shall be cause for rejection.

4.4.1.2 Depth setting mechanism (for deep firing).

4.4.1.2.1 Valve opening pressure test (1000 foot setting).-- With the depth setting pointer turned to the 1000 foot mark, the assembly shall be placed in the test fixture, Drawing No. 423552, referenced on Sketch No. 109321, and subjected to water pressure. The applied water pressure shall be increased at the rate of approximately 11.0 pounds per square inch per second until the valve opens. Valve opening at this depth setting shall be within the range of 415 to 445 pounds per square inch. If the valve opening pressure is not within the range noted, the depth setting mechanism shall be recalibrated as specified in 3.6.1. Chattering or vibrating of the valve at the point of opening within the required range as noted shall be cause for rejection.

4.4.1.2.2 Leakage test (1000 foot setting). -- Leakage may sometimes occur through the gasket at low pressure and not at high pressure. Therefore, examination for this defect shall be made while pressure is rising from 0 to the valve opening pressure as specified in 4.4.1.2.1. Leakage, at any pressure below valve opening pressure, in excess of five cubic inches (82cc) per minute shall be cause for rejection.

4.4.1.2.3 Valve opening pressure test (400 foot setting).-- Repeat the test specified in 4.4.1.2.1 with the depth setting pointer turned to the 400 foot mark. The valve opening pressure range shall be 150 to 180 pounds per square inch. Subassemblies which fail to meet this requirement shall be rejected.

4.4.1.2.4 Valve closing pressure test (400 foot setting).-- As specified in 3.5.4, the valve closing pressure at the 400 foot setting shall be within 46 to 90 pounds per square inch of the opening pressure. The closing pressure shall be obtained after the valve opens by a gradual reduction of applied pressure until the valve snaps shut. Subassemblies which fail to meet this requirement shall be rejected.

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4.4.1.2.5 Leakage test (400 foot setting). -- After the valve is completely closed in accordance with the requirements for the 400 foot setting, the pressure shall be raised to 120 pounds per square inch and held while an examination is made for leakage. An amount of leakage not in excess of 5 cubic inches (82cc) in one minute shall be considered passable. Leakage greater than this amount shall be cause for rejection.

4.4.2 Leakage test (700 foot setting). -- During the second valve operating cycle specified in 3.6.1.4, the mechanisms shall be inspected for leakage at a pressure of 170 pounds per square inch. Leakage in excess of five cubic inches (82cc) per minute shall be cause for rejection. The 170 pounds per square inch pressure shall be attained by increasing the pressure from 140 pounds per square inch.

4.4.3 "300" index pointer setting test. -- The depth setting mechanism of the assembled pistol shall be set at "0-to-300" and the pistol tested with compressed air for firing at the "300" index pointer setting. Failure to pass this test shall be cause for rejection.

4.4.4 Test of calibration. -- Rotate the index pointer clockwise to "SAFE", then counterclockwise to "30", and clockwise again to "100". With the depth setting mechanism for deep firing set at "0-to-300", determine with compressed air, the firing pressure corresponding to the "100" setting. If the pressure is not between 43 and 46 pounds per square inch, recalibrate the pistol as specified in 3.6.2.

4.4.5 Maximum pressure tests. -- For maximum pressure tests, the pistol shall be mounted on the test fixture shown on Drawing Nos. 423552, 437961, and 423554, referenced on Sketch No. 109321. The test fixture shall be adequately supplied with water and equipped with a suitable means, such as compressed air for smoothly controlling the water pressure between 0 and 500 pounds per square inch. Provision shall be made to relieve any air pockets in the fixture after the pistol is secured in place. A gauge shall be attached to the fixture. To avoid the effect of any drop in line pressure when water is flowing into the pistol, the gauge shall be attached in such a way as to record hydrostatic pressure at the top of the pistol.

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4.4.5.1 The following procedures shall apply for all the tests specified in 3.9.

4.4.5.1.1 Apply water pressure at the rate of approximately 11 pounds per square inch per second until the pistol fires.

4.4.5.1.2 At the point of firing (not point of valve opening) note the gauge pressure. The highest pressure attained by the gauge before firing occurs shall be considered as the firing pressure.

4.4.5.2 In addition to those specified in 4.4.5.1.1 and 4.4.5.1.2, the following procedures shall apply for the tests specified in 3.9.3 and 3.9.4. For the requirement of paragraph 3.9.4, sampling shall be in accordance with MIL-STD-105 and with the following table using an AQL of 1.0 percent:

LOT SIZE	SAMPLE SIZE CODE LETTER
1300 and under	F
1301 and over	I

4.4.5.2.1 After test for firing at each setting, empty water from the pistol by removing the subassembly shown on Drawing No. 432885.

4.4.5.2.2 Before replacing subassembly, check to insure that the gasket is properly installed.

4.4.5.3 Pistols conforming to the requirements of the final maximum pressure test shall be disassembled and the parts blown out with air. All parts except subassembly Drawing No. 432885 shall then be placed in an oven at 220 degrees F for at least four hours to remove all moisture. Damaged parts shall be replaced. Subassembly Drawing No. 432885 shall be dried by blowing out with compressed air. The reassembled pistols, calibrated, tested, and sealed as specified in 3.5, 3.6, 3.7, and 3.8, will be considered acceptable as part of a contract lot.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -- Each depth charge pistol shall be treated with rust preventive compound as specified in 3.3 and packaged in a corrugated paper carton constructed as shown on Drawing Nos. 421087, 421088, and 421163, referenced on Sketch No. 109321.

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5.1.1 Special attention shall be given to the preparation of pistols to withstand long periods of dry storage in the Contractor's packaging at ordinary atmospheric conditions of temperature and humidity.

5.1.2 Pistols shall be delivered in the cocked condition, with plain type inlet valve covers in place, with the index pointer set on "30" and with the depth setting mechanism for deep firing set at "500". One knobbed type inlet valve cover, one wrench stop, and one safe setting lock for each pistol shall be included in each carton.

5.2 Packing. -- Ten depth charge pistols, packaged as specified in 5.1, shall be packed in a wooden box constructed as shown on Drawing No. 433982, referenced on Sketch No. 109321.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES. -- There are none applicable.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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